Wireless and Mobile Networking (02.11.2026)
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MASTER SOLUTIONS KEY: VERSION A & VERSION B

For Grading Evaluation Reference Only

1. Version A Evaluation Reference

Answer Key Matrix:

Question	Correct Choice	Core Topic	Lecture Source Reference
Q1	3rd Option	Guided Media	Week 1: Section 1.1 (A.a) — Page 1
Q2	2nd Option	5G Core Architecture	Week 2: Section 2.1.3 — Page 3
Q3	4th Option	Digital Modulation	Week 3: Section 3.1 & 3.3 — Pages 1, 2
Q4	1st Option	OFDM Subcarrier Alignment	Week 3: Section 3.4 — Page 3
Q5	2nd Option	MAC Layer Anomalies	Week 4: Section 4.3.2 — Page 3

2. Version B Evaluation Reference

Answer Key Matrix:

Question	Correct Choice	Core Topic	Lecture Source Reference
Q1	2nd Option	OFDM Subcarrier Alignment	Week 3: Section 3.4 — Page 3
Q2	1st Option	Guided Media	Week 1: Section 1.1 (A.a) — Page 1
Q3	3rd Option	MAC Layer Anomalies	Week 4: Section 4.3.2 — Page 3
Q4	4th Option	5G Core Architecture	Week 2: Section 2.1.3 — Page 3
Q5	1st Option	Digital Modulation	Week 3: Section 3.1 & 3.3 — Pages 1, 2

3. Technical Solution Rationales (Both Versions)

Guided Media (Twisted Pair Cables): Helical twisting cancels out opposing magnetic fields generated by the copper lines. This isolates individual paths and prevents crosstalk noise leaking into nearby channels [Week 1, Page 1].

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5G Core Architecture: The 5G Core departs from rigid legacy hardware elements by decomposing functions into cloud-native software containers called Network Functions (NFs) running standard RESTful APIs over an internal service bus [Week 2, Page 3].

Digital Modulation Trade-offs: Higher modulation orders like 64-QAM expand system capacity by packing more payload bits into every symbol duration. However, because constellation indices sit closer on the I-Q plane, the receiver demands a much higher Signal-to-Noise Ratio (SNR) to clear interference margins [Week 3, Page 2].

OFDM Subcarrier Alignment: OFDM divides channels into tightly nested parallel subcarriers. Overlap is safe because the subcarrier spacing guarantees that when one subcarrier hits its peak amplitude, all adjacent subcarriers pass perfectly through their null zero-crossings, neutralizing Inter-Carrier Interference (ICI) [Week 3, Page 3].

Exposed Terminal Anomalies: Exposed terminals interpret local open airspace as busy because they overhear a transmission from a nearby speaker. They needlessly hold back their frames, failing to realize that their target receiver is located well beyond the current sender's physical footprint [Week 4, Page 3].

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